

SPRINT 2026

Project Description

PROJECT NAME: PLANETIA UTOPIA

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GROUP: 9

SUMMARY:

A RasbotV2-based mobile robot (Raspberry Pi, Mecanum drivetrain, Intel RealSense D405, ultrasonic and IR sensors) that traverses a room and builds a merged 360° point cloud at each sampled location. Controlled via a Python API with two modes: a manual mode where the user drives the robot in real time from a keyboard (WASD keys for omnidirectional movement and in-place rotation, with a single key triggering a 360° capture), and an autonomous mode that follows a dark-tape path with cross-mark stop markers. Built with OpenCV, NumPy, and pyrealsense2.

WORK CONTENT:

1. Bring up the Raspberry Pi 5: install the OS, enable the I2C and camera interfaces, and set up the Python environment with the required libraries.
2. Assemble the RasbotV2 robot: mount the motors, Mecanum wheels, RealSense D405, ultrasonic, and IR line sensors, and wire everything to the I2C motor driver and the Raspberry Pi 5.
3. Set up the network: connect the Raspberry Pi 5 to a 2.4 GHz Wi-Fi router and enable remote access over SSH and VNC.
4. Build a Python API abstracting motion, sensing, and tilt.
5. Drive the robot from the keyboard: WASD teleop over the Python API for omnidirectional movement and in-place rotation.
6. Stereo depth and 360° merging: compute disparity from stereo pairs, back-project to a metric point cloud, and merge per-orientation clouds at each location into a single 360° cloud (ICP-refined).
7. Implement Mode 1: keyboard-driven manual control (WASD for omnidirectional movement and in-place rotation, plus a capture key).
8. Implement Mode 2: autonomous IR line following with cross-mark stop markers.
9. Integration: build a top-level module that wires the API, depth pipeline, capture routine, and both navigation modes into a single runnable application.
10. Validation and testing: preparing the room and run both modes across multiple capture locations and check navigation, per-location 360° clouds, and the merged room reconstruction.

PROJECT EXPECTED OUTPUT (WITH A DIAGRAM IF RELEVANT):

A working RasbotV2 robot, a Python hardware-abstraction API, a stereo depth + 360° merging module, and both manual and autonomous-line-following applications. Output artifacts are per-capture and merged .ply point clouds and a multi-location room reconstruction, delivered with documentation and a short demo video of each mode.



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SCHEDULE:

Fill the table below with all major tasks, milestones, and deliverables of the project. Describe each task using a paragraph (references to previous sections can be used). As a general rule, at least one milestone for each day of the SPRINT is expected.

Task/ Milestone/ Deliverable	Name of Task/Milestone/Deliverable	Expected Date and Time to be Completed	Description	Possible Dependencies and/or additional Comments
Task	1. Pi 5 bring-up	Sunday	OS install, enable I2C and camera, set up Python environment.	—
Task	2. Robot assembly	Sunday	Mount motors, Mecanum wheels, RealSense D405, ultrasonic, and IR line sensors; wire to driver/Pi.	Depends on 1
Task	3. Network setup	Sunday	Connect Pi 5 to 2.4 GHz Wi-Fi router; enable SSH and VNC.	Depends on 1
Milestone	Robot communication established	Sunday	Remote connection and communication with the robot verified: SSH and VNC sessions to the Pi 5 from a host machine over the 2.4 GHz Wi-Fi link are stable and ready for development and monitoring.	Depends on 3
Task	4. Python API	Sunday	Hardware-abstraction API for motion, sensing, and tilt.	Depends on 1, 2, 3
Task	5. Keyboard driving	Monday	WASD teleop over the Python API for omnidirectional movement and in-place rotation.	Depends on 4
Task	6. Stereo depth + 360° merging	Monday	Compute disparity from stereo pairs, back-project to a metric point cloud, and merge per-orientation clouds at each location into a single 360° cloud (ICP-refined).	Depends on 4
Task	7. Mode 1 — Manual control	Monday-Tuesday	Keyboard-driven WASD teleop with a capture-key trigger for the autonomous routine.	Depends on 4, 6
Task	8. Mode 2 — Line following	Monday-Tuesday	Autonomous IR line following with cross-mark stop markers triggering the capture routine.	Depends on 4, 6
Task	9. Integration	Wednesday	Top-level module wiring API, depth pipeline, capture routine, and both navigation modes.	Depends on 7, 8
Deliverable	10. Validation & testing	Thursday	Run both modes across multiple capture locations; verify navigation, 360° clouds, room reconstruction.	Depends on 9

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GANTT:

SPRINT Project Schedule - Gantt Chart

